

LA HW

Included exercises: 1, 3, 7, 9, 13, 15, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37

Exercises 1-10

In Exercises 1–10, determine $[T]_{\mathcal{B}}$ for each linear operator T and basis $\mathcal{B}$.

Original Exercise 1

BASIS $\mathcal{B}$.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + x_2 \\ x_1 - x_2 \end{bmatrix} \text{ and } \mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Original Exercise 3

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ x_1 + x_2 \end{bmatrix} \text{ and } \mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$$

Original Exercise 7

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 4x_2 \\ x_1 + 2x_3 \\ -2x_2 + 3x_3 \end{bmatrix} \text{ and}$$

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} \right\}$$

Original Exercise 9

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 \\ x_2 - x_3 \\ x_1 + 2x_4 \\ x_2 - x_3 + 3x_4 \end{bmatrix} \text{ and}$$

Exercises 1-25

In Exercises 1–25, determine whether the statements are true or false.

Original Exercise 13

$$[T]_{\mathcal{B}} = \begin{bmatrix} -2 & -1 \\ 1 & 3 \end{bmatrix} \text{ and } \mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 5 \end{bmatrix} \right\}$$

Original Exercise 15

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & -3 \\ -2 & 1 & 2 \\ -1 & 1 & 1 \end{bmatrix} \text{ and}$$

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Exercises 19-38

In Exercises 19–38, determine whether the statements are true or false.

Original Exercise 19

ments are true or false.

A linear transformation $T: \mathcal{R}^n \rightarrow \mathcal{R}^m$ that is one-to-one

Original Exercise 21

If T is a linear operator on $\mathcal{R}^n$ and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is a basis for $\mathcal{R}^n$, then column j of $[T]_{\mathcal{B}}$ is the $\mathcal{B}$ -coordinate

Original Exercise 23

$$[T(\mathbf{b}_1) \ T(\mathbf{b}_2) \ \cdots \ T(\mathbf{b}_n)].$$

If $\mathcal{E}$ is the standard basis for $\mathcal{R}^n$, then $[T]_{\mathcal{E}}$ is the standard

Original Exercise 25

the standard matrix of T , then $[T]_{\mathcal{B}} = B^{-1}A$.

If T is a linear operator on $\mathcal{R}^n$, $\mathcal{B}$ is a basis for $\mathcal{R}^n$, B is the matrix whose columns are the vectors in $\mathcal{B}$, and A is

Original Exercise 27

If T is a reflection of $\mathcal{R}^2$ about a line L , then $T(\mathbf{v}) = -\mathbf{v}$

Original Exercise 29

If T is a reflection of $\mathcal{R}^2$ about a line, then there exists a basis $\mathcal{B}$ for $\mathcal{R}^2$ such that $[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Original Exercise 31

a basis $\mathcal{B}$ for $\mathcal{R}^2$ such that $[T]_{\mathcal{B}} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

If T , $\mathcal{B}$, and L are as in Exercise 30, then $\mathcal{B}$ consists of

Original Exercise 33

An $n \times n$ matrix A is similar to an $n \times n$ matrix B if B

Original Exercise 35

The only matrix that is similar to the $n \times n$ zero matrix

Original Exercise 37

If T is a linear operator on $\mathcal{R}^n$ and $\mathcal{B}$ is a basis for $\mathcal{R}^n$,